

Tide retrospective: the sector lower bound

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

2026-08-09

1 Setting

Second tide of the germbij arc, following `2026-08-08-19-11-tide-germbij-pencil`, which landed the pencil identity for pairs of potentials. The identifiability theorem of the germbij note (“What expectation values know about the loss landscape”, SRI, August 2026, Theorem 7.3) needs one more quantitative ingredient: a polynomial-in- t lower bound for $\int a^2 \psi e^{-tK}$ when $K \geq 0$ vanishes at a point with at most quadratic growth and a has finite vanishing order there. This tide formalises that bound in one dimension, with the analytic-germ input factored into the hypothesis $c w^m \leq |a(w)|$ on $[0, r_0]$ (which is what analyticity supplies after shrinking the neighborhood and choosing the side of the sign). The candidate was drafted and deliberated during the previous tide’s consult (candidate B there), so this tide proceeded on the deliberation of record without a fresh consult.

2 The thing formalised

Two statements in `Laplace/Sector.lean`, both sorry-free. The scale-window lemma: for $0 < u$, $2u \leq r_0$, $0 \leq c$, $0 \leq C_0$, $0 \leq t$, with $K \leq C_0 w^2$ and $c w^m \leq |a(w)|$ on $[0, r_0]$ and the integrand continuous on the window,

$$u \cdot c^2 u^{2m} e^{-4C_0 t u^2} \leq \int_{[u, 2u]} a(w)^2 e^{-tK(w)} dw.$$

The Laplace-scale corollary (`sector_lower_bound`): substituting $u = (\sqrt{t})^{-1}$, under $4 \leq r_0^2 t$,

$$c^2 e^{-4C_0 t} t^{-m-1/2} \leq \int_{[(\sqrt{t})^{-1}, 2(\sqrt{t})^{-1}]} a(w)^2 e^{-tK(w)} dw,$$

with the exponent as a real power $t^{-(m:\mathbb{R})-1/2}$. This is germbij Lemma 7.2 in dimension one. The numerical check (previous tide’s log): at $t = 100$, $K = w^2$, $a = w^2$, $r_0 = 1$, the integral is 4.6×10^{-6} against the bound 1.8×10^{-7} .

3 Proof strategy

The window lemma is a constant lower bound integrated over an interval: on $[u, 2u]$ the amplitude factor satisfies $a^2 \geq c^2 u^{2m}$ (monotonicity of powers plus $c w^m \leq |a|$ and $|a|^2 = a^2$) and the Boltzmann factor satisfies $e^{-tK} \geq e^{-4C_0 t u^2}$ (from $K \leq C_0 w^2 \leq 4C_0 u^2$), so the pointwise product bound holds; the set-integral monotonicity lemma against a constant¹ finishes. The corollary is

¹`MeasureTheory.setIntegral_ge_of_const_le`, with `ContinuousOn.integrableOn_compact` supplying integrability and `Real.volume_real.Icc_of_le` evaluating the window length to u .

pure algebra of the substitution $u = (\sqrt{t})^{-1}$: positivity of t from $4 \leq r_0^2 t$ (by contradiction), $2 \leq r_0 \sqrt{t}$ from squaring, $tu^2 = 1$, $u^{2m} = (t^m)^{-1}$, and the real-power identity $t^{-m-1/2} = (t^m)^{-1}u$ via `Real.rpow_add`, `Real.rpow_neg`, `Real.rpow_natCast`, and `Real.sqrt_eq_rpow`.

4 Roadblocks and resolutions

None of substance: the file compiled on the first serious attempt, with two lint-level fixes. One mathematical simplification surfaced by the linter: the deliberated statement carried the hypothesis $0 \leq K$, but the proof never uses it (only the upper bound on K enters a lower bound on the Boltzmann factor), so the landed statement is slightly stronger than the deliberated one. The `rpow/sqrt` bookkeeping that the deliberation flagged as the main Lean cost was real but contained: a five-rewrite chain covers it. The `nlinarith` calls needed the usual explicit product hints (`sq_nonneg` of the right expressions).

5 What was Mathlib, what was new

All Mathlib lookup: set-integral monotonicity, interval volume, power monotonicity², `Real.sq_sqrt` and the `rpow` API. New content: the two statements and the choice of hypothesis factoring (the analytic input as $cw^m \leq |a|$), which is what makes the statement elementary while remaining exactly the `germbij` Lemma 7.2 shape.

6 Lessons

When a deliberated statement carries a hypothesis the proof does not use, drop it and note the strengthening; the linter catches this reliably. The window-then-substitute decomposition (prove the bound on $[u, 2u]$ with u free, then specialise $u = (\sqrt{t})^{-1}$) cleanly separates measure theory from `rpow` algebra and is worth reusing whenever a Laplace-scale bound is needed.

7 Follow-ups

The composite tide: `germbij` Theorem 7.3 in dimension one, combining the integrated pencil identity and comparison lemma (landed in the pencil tide) with the sector bound of this tide, into: if $L_1, L_2 \geq 0$ vanish at 0 with $L_1 + L_2 \leq C_0 w^2$ and $g = L_2 - L_1$ satisfies $cw^m \leq |g|$ on $[0, r_0]$, then for a bump $\psi \geq 0$ equal to 1 near 0, the observable $\phi = g\psi$ gives $\int \phi (e^{-tL_1} - e^{-tL_2}) \geq c^2 e^{-4C_0} t^{1-m-1/2}$ for $t \geq 4/r_0^2$: the quantity that contradicts $o(t^{-\infty})$ and proves 1D identifiability. The glue steps are restricting the integral to the window (nonnegativity of the integrand via `exp_pencil_ge`) and the pencil average.

²`pow_le_pow_left`.